

Eliane Juang

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EDUCATION

University of Pennsylvania

Philadelphia, PA

BSE in Mechanical Engineering | Minor in Engineering Entrepreneurship | GPA: 3.99/4.0

May 2029

Relevant Coursework: Statics and Strength of Materials, Introduction to Thermal-Fluids Engineering, Machine Design and Manufacturing, Differential Equations, Linear Algebra

EXPERIENCE

GRASP Sung Robotics Lab

May 2026 – Present

Undergraduate Researcher | *Liebau Underwater Soft Robot & Mitral Valve Projects*

Philadelphia, PA

- Designed 3 actuator variants (electromagnetic tapping actuator, scotch-yoke tapping actuator, silicone-molded concentric pinching actuator) for a Liebau-powered underwater soft robot; theoretical calculations show the optimized electromagnetic design cuts power consumption by 50%
- Evaluated a scotch-yoke actuator, measuring flow output at ~25% of the electromagnetic design
- Built 6 iterative prototypes of the mitral valve mechanism, scaling from 10x to target size; calculated the coil design for an axial flux motor from first principles

Airalux Medical

Aug 2026 – Present

Engineering Intern | *Therapeutic Spirometer*

Philadelphia, PA

- Analyzed syringe-pull trials on 5 device prototypes to characterize airflow rate vs. pressure differential across environmental conditions, evaluating accuracy under simulated clinical settings
- Modeled new components in CAD to improve device usability and features; soldered pins on the prototype PCB

ENGINEERING TEAMS

Penn Hyperloop | Mechanical Team Member, *Tunnel Boring Machine*

Sep 2026 – Present

- Redesigning the cutterhead to meet the target specifications of 2700 N-m and 10 RPM; performing buckling, shear stress, and FEA during design
- Manufacturing components by welding, in preparation for the Boring Company's annual Not-a-Boring Competition

Penn Aerospace Club | Payload Team Member, *High-Altitude Balloon*

Sep 2025 – Present

- Redesigned energy-absorbing components after post-launch structural failure analysis; used FEA stress plots to compare candidate designs and select the strongest structure
- Iterated payload components in CAD to meet the 4-pound payload weight limit; payload reached 30,000+ ft

Penn Engineering Without Borders | Project Member, *Novick Urban Farms Project*

Feb 2026 – Present

- Developed and assembled 4 vertical planter box systems compatible with existing agricultural equipment for year-round crop production; implemented blanket liners and a funnel system for wastewater filtering

Access Engineering | Mechanical Engineering Teaching Assistant

Feb 2026 – Present

- Teach Onshape CAD, laser cutting, and 3D printing to 30+ Philadelphia high schoolers over 6 sessions per semester; mentor students in designing truss bridges, applying tension and compression analysis

PROJECTS

Stirling Engine | *MEAM 2010*

Aug 2026 – Dec 2026 (expected)

- Machining a Stirling engine from raw stock via milling, CNC, bandsaw, and tapping, holding tolerances of ± 0.005 in

Force-Sensing LED Game | *ENGR 1050*

Dec 2025

- Programmed a Raspberry Pi Pico to map force-sensing resistor input to proportional LED output, comparing the force applied against random target values via a button-triggered capture

SKILLS

CAD: SolidWorks (Certified SolidWorks Associate), Onshape

Manufacturing: CNC, Milling, 3D Printing, Laser Cutting, Molding, Welding

Programming: Java, Python, MATLAB

Electronics: Arduino, ESP32, Raspberry Pi Pico